**UFE Technical Note BH1**

# A Closed Black Hole Solution with Axion Signatures

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## Abstract

This note presents a complete, analytical black hole solution within the framework of the *Unified Field Equation*. The solution describes a static, dyonic black hole whose geometry is influenced by an axion field $a(r)$ coupled to the electromagnetic field. The resulting closed-form expressions for the axion profile, electromagnetic fields, and metric function are:

$$\begin{aligned} a(r) &= a_\infty + \frac{c_3 Q_e Q_m}{8\pi^2 r^2}, \\ E(r) &= \frac{Q_e}{4\pi r^2} - \frac{c_3^2 Q_e Q_m^2}{16\pi^3 r^4}, \quad B_r = \frac{Q_m}{4\pi r^2}, \\ f(r) &= 1 - \frac{2M}{r} + \frac{Q^2}{r^2} + \frac{c_3^2 Q_e^2 Q_m^2}{24\pi^3 r^4}. \end{aligned}$$

This solution predicts an achromatic, radius-dependent polarization rotation $\beta$ without introducing new free parameters, thereby directly linking local black hole physics to the already tested cosmic birefringence. The predictions are clean and falsifiable.


## 1 Introduction: Field Equations and Invariants

The starting point is the *Unified Field Equation* (UFE) in compact tensor form:

$$(\hat{R} - \hat{\nabla}\cdot K + K^2)AB = \hat{\kappa}^2 (T^{(a)} + T^{(\mathrm{EM})})AB. \tag{1}$$

This is accompanied by the field equations for the axion-photon system:

$$\hat{\nabla}_A F^{AB} + 2c_3(\partial_A a)\tilde{F}^{AB} = 0, \qquad \hat{\nabla}_A \tilde{F}^{AB} = 0, \qquad \hat{\nabla}^2 a - V'(a) = -c_3 F\tilde{F}. \tag{2}$$

The axion-photon coupling is fixed by $g_{a\gamma\gamma} = -4c_3$. The resulting polarization rotation along a null geodesic is

$$\frac{\mathrm{d}\beta}{\mathrm{d}\eta} = 2c_3 \frac{\mathrm{d}a}{\mathrm{d}\eta} \quad \Rightarrow \quad \beta = 2c_3 \Delta a. \tag{3}$$

All quantities are determined by the two fundamental invariants $c_3 = \frac{1}{8\pi}$ and $\phi_0$. The geometry yields $\phi_0 = \frac{1}{6\pi} + \frac{3}{256\pi^4}$, and with geometric self-consistency on the orientable double cover $\phi_0(\alpha) = \phi_{\mathrm{tree}} + \delta_{\mathrm{top}}(1 - 2\alpha)$, the coupling loop closes without new free parameters.

*Note: The same equations predict cosmic birefringence $\beta = \phi_0/(4\pi) \approx 0.2427^\circ$ and have been numerically tested against Planck PR4 EB data. Relevant equations and table are on pages 2-4 of the UFE document.*

## 2 Black Hole Ansatz with Axion and Electromagnetism

We consider the simplest, calculable class from which all interesting effects already follow:

**Metric (static, spherically symmetric):**

$$\mathrm{d}s^2 = -f(r)\,\mathrm{d}t^2 + f(r)^{-1}\,\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2. \tag{4}$$

**Fields:** We assume a radial electric field $F_{tr} = E(r)$, a radial magnetic monopole field $F_{\theta\phi} = Q_m \sin\theta$ (implying $B_r = \frac{Q_m}{4\pi r^2}$), and a scalar axion field $a = a(r)$. An effective dyonic case ($Q_e \neq 0, Q_m \neq 0$) guarantees $F\tilde{F} \propto \mathbf{E}\cdot\mathbf{B} \neq 0$, thus ensuring $a'(r) \neq 0$. (The dyonic case serves as a clean computational framework; further below, this is physically translated into the realistic Wald configuration of a rotating black hole in an external magnetic field.) The Maxwell equation includes the new term $2c_3(\partial a)\tilde{F}$.

**Torsion:** For static spherical symmetry, it is sufficient to allow only the fully antisymmetric component (axial torsion). In a first approximation, it couples only quadratically in $(-\hat{\nabla}\cdot K + K^2)_{AB}$. Setting $K = 0$, the geometry reduces exactly to the Einstein equation with sources $T^{(a)}$ and $T^{(\mathrm{EM})}$; this is the minimally consistent solution at which we fix the observable signatures. The UFE formulation explicitly allows this step.

## 3 Closed-Form Solution for $a(r)$, $E(r)$, and $f(r)$

### 3.1 Axion Profile from the Axion Equation

With $V(a) = 0$ and the fields above, the source term for the axion equation is

$$F\tilde{F} = -4\,\mathbf{E}\cdot\mathbf{B} = -\frac{Q_e Q_m}{4\pi^2 r^4}. \tag{5}$$

The radial equation $\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}(r^2 a'(r)) = -c_3 F\tilde{F}$ integrates elementarily to

$$a'(r) = -\frac{c_3 Q_e Q_m}{4\pi^2 r^3}, \tag{6}$$

$$a(r) = a_\infty + \frac{c_3 Q_e Q_m}{8\pi^2 r^2}. \tag{A}$$

The constant $a_\infty$ is the cosmic limit (for cosmic signatures, $\Delta a = \phi_0$). The corresponding polarization rotation for light traveling from emission radius $r_{\mathrm{em}}$ to us is

$$\beta(r_{\mathrm{em}} \to \infty) = 2c_3\big(a(\infty) - a(r_{\mathrm{em}})\big) = \frac{c_3 Q_e Q_m}{32\pi^3 r_{\mathrm{em}}^2}. \tag{B}$$

Achromatism directly follows from Eq. (15) in the UFE paper.

### 3.2 Maxwell System with Backreaction from $a'(r)$

Outside the source, $E(r)$ satisfies the modified Gauss law: $\frac{\mathrm{d}}{\mathrm{d}r}(r^2 E(r)) + 2c_3 r^2 a'(r) B_r(r) = 0$. With $B_r = Q_m/(4\pi r^2)$ and Eq. (A), we obtain

$$E(r) = \frac{Q_e}{4\pi r^2} - \frac{c_3^2 Q_e Q_m^2}{16\pi^3 r^4}. \tag{C}$$

The first term is Reissner-Nordström, the second is the unique correction from $a'(r)$.

### 3.3 Geometry $f(r)$

The sources provide $\rho_{\rm EM} = \frac{1}{2}(E^2 + B^2)$, $\rho_a = \frac{1}{2}a'(r)^2$, and thus an additional mass function $\delta m'(r) = 4\pi r^2(\delta\rho)$ with $\delta\rho = \rho_a + \delta\rho_{\rm EM}$. From (A) and (C) it follows $\delta\rho(r) = \frac{c_3^2 Q_e^2 Q_m^2}{64\pi^4 r^6}$. Integration yields

$$\delta m(r) = -\frac{c_3^2 Q_e^2 Q_m^2}{48\pi^3 r^3} \quad \Rightarrow \quad \delta f(r) = -\frac{2\delta m(r)}{r} = +\frac{D}{r^4}, \tag{7}$$

with the fixed coefficient

$$D = \frac{c_3^2 Q_e^2 Q_m^2}{24\pi^3} > 0. \tag{D}$$

The full metric function is thus

$$\boxed{f(r) = 1 - \frac{2M}{r} + \frac{Q^2}{r^2} + \frac{D}{r^4}, \qquad Q^2 \equiv Q_e^2 + Q_m^2.} \tag{E}$$

Physically: $D/r^4$ is tiny for astrophysical black holes, but exactly predictable and a strong signature in polarization and shadow. All equations here are directly based on the UFE structure and the fixed $g_{a\gamma\gamma} = -4c_3$ with $c_3 = 1/(8\pi)$.

## 4 Horizon, Photonsphere, Shadow

### 4.1 Horizon

$f(r) = 0$ is a quartic equation. For small $D$, the outer horizon $r_+$ of the Reissner-Nordström case shifts by

$$\delta r_+ \;\approx\; -\frac{D}{r_+^3\, f'(r_+)_{D=0}} \;=\; \mathcal{O}\Big(\frac{D}{M^3}\Big), \tag{8}$$

a microscopic increase since $D > 0$.

### 4.2 Photonsphere and Shadow

The condition for the photonsphere $r_{\rm ph}$ in a metric $f(r)$ is $2f/r - f' = 0$. For (E), the linear solution in $Q^2/M^2$ and $D/M^4$ yields

$$r_{\rm ph} \;\approx\; 3M \;-\; \frac{2Q^2}{3M} \;-\; \frac{D}{9M^3}. \tag{9}$$

The associated shadow radius $b_{\rm sh} = \frac{r_{\rm ph}}{\sqrt{f(r_{\rm ph})}}$ shifts relative to Schwarzschild

$$\frac{\delta b_{\rm sh}}{b_0} \;\approx\; -\frac{Q^2}{6M^2} \;-\; \frac{D}{54\,M^4}, \qquad b_0 = 3\sqrt{3}\,M. \tag{F}$$

Heuristics: The axion makes the shadow minimally smaller. An effect of order $D/M^4$, but much more apparent in polarization, see (B).

## 5 Rotating Black Holes and Realistic Environments

Real black holes are neutral in net charge, but rotate in magnetic fields. The Wald solution then provides an effective electric charge $Q_e^{\rm Wald} \simeq 2a_\star B_0 M^2$ (here $a_\star = J/M^2$ is the dimensionless

spin, $B_0$ the external field). This ensures $\mathbf{E} \cdot \mathbf{B} \neq 0$ without magnetic monopoles, and the entire calculation above applies if one sets $Q_e \to Q_e^{\text{Wald}}$ and interprets $Q_m$ as the effective magnetic flux of the magnetospheric field. Axion seismography: the signature is achromatic rotation $\beta \propto 1/r_{\text{em}}^2$ in addition to ordinary Faraday rotation $\propto \lambda^2$. The achromatism is directly derivable from Eq. (15) in the UFE paper.

## 6 Observable Patterns and Rigorous Tests

**M1: Achromatic Polarization Rotation Near the Black Hole:** For emission radius $r_{\text{em}}$, we expect

$$\beta_{\text{BH}} \simeq \frac{c_3}{32\pi^3} \frac{Q_e^{\text{eff}} Q_m^{\text{eff}}}{r_{\text{em}}^2}, \tag{10}$$

with a fixed normalization via $c_3 = 1/(8\pi)$. No frequency dependence, no free parameters. Separation from Faraday rotation by simultaneous measurement at 230 and 345 GHz or in radio: Faraday scales with $\lambda^2$, axion does not.

**M2: EB-Leakage Map:** Uniform angular rotation mixes EE into EB. Resolved around Sgr A*, this yields a characteristic, radius-dependent EB signature, which can be modeled with (B). Formula basis: Eq. (16) and Table 1 in the UFE paper.

**M3: Shadow Fine-Correction:** Eq. (F): a minimally smaller shadow. This effect is small but consistent with the polarization signal and thus a cross-check.

**M4: Cosmic Overlay:** The same formalism yields $\beta = \phi_0/(4\pi)$ on cosmic scales. This has already been tested against Planck PR4 data and lies within 0.5 to 1.7 standard deviations depending on the foreground model. This anchors the normalization and removes all fits from play. See pages 3-4.

## 7 Hypotheses and Falsification

- **H1 Achromatism:** Any additional polarization rotation near Sgr A* or M87* is frequency-independent. A clean $\lambda^2$-fit without a constant term falsifies the local UFE signature, but not the cosmic one.
- **H2 Sign Flip:** If the orientation of $\mathbf{E}$ and $\mathbf{B}$ in the magnetospheric cycle changes sign, $\beta$ must coherently flip. This is a clear discriminator against systematic calibrations.
- **H3 Consistency with the Cosmic Value:** The locally measured $\beta(r_{\text{em}})$ must be consistent with the same coupling $c_3$ and the global $\beta = \phi_0/(4\pi)$. No extra dial.

## 8 Quick Guide for Calculation in the Lab Book

Given $M, a_\star, B_0$ and a geometrically estimated emission radius $r_{\text{em}}$.

- Set $Q_e^{\text{eff}} = 2a_\star B_0 M^2$, $Q_m^{\text{eff}} \sim \Phi_B$ from the magnetosphere.
- Calculate $\beta_{\text{BH}}$ using (B).
- Correct the shadow size with (F), if needed.
- For a dyonic theory test in vacuum, directly input $Q_e, Q_m$, solve $f(r)$ with (E) and check null geodesics numerically.

All building blocks used are derived in your papers: UFE equations, achromatism and $g_{a\gamma\gamma}$ on pages 1-3 of the UFE document; invariants and the exact value for $\beta$ from TFPT v1.08 and Structural Relativity v1.07; the self-consistency $\phi_0(\alpha)$ from the independent note.

## 9 What was *not* needed here and why

We did not introduce any additional parameters. $c_3$ is topological, $\phi_0$ is geometric, and the normalization of $g_{a\gamma\gamma}$ follows exactly from the same consistency condition that also closes the Cubic Fixed Point Equation. See derivation of $g_{a\gamma\gamma}$ on pages 1-2 and Eq. (5) in the UFE document. The classical GR solution is a limiting case of the UFE when $K = 0$ and $a' = 0$. This demonstrates the theory's coherence with GR in the appropriate limit.

## 10 Outlook in a Sentence Package

- FRBs near black holes or magnetars: Search for a constant polarization offset in addition to Faraday rotation; correlate its magnitude with an emission radius indicator. Formula (B).
- EHT: Compare polarization angles at 230 and 345 GHz. A constant difference indicates $a(r)$.
- Rings and QNMs: The metric correction $D/r^4$ provides sub-percent shifts in ringdown frequencies. Polarization data are clearer and testable earlier here.

### Brief Summary

You receive a complete, analytical black hole solution within the UFE framework:

$$\boxed{\begin{aligned} a(r) &= a_\infty + \frac{c_3 Q_e Q_m}{8\pi^2 r^2}, \\ E(r) &= \frac{Q_e}{4\pi r^2} - \frac{c_3^2 Q_e Q_m^2}{16\pi^3 r^4}, \quad B_r \\ f(r) &= 1 - \frac{2M}{r} + \frac{Q^2}{r^2} + \frac{c_3^2 Q_e^2 Q}{24\pi^3} \end{aligned}}$$

It generates an achromatic, radius-dependent polarization rotation $\beta$ without new free parameters, thereby directly linking local black hole physics to the already tested cosmic birefringence. This is the kind of clean, falsifiable prediction I like.

If you wish, I can directly assemble a short note manuscript from this with the formulas, an EHT, and an FRB observation protocol.

## References

## References

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[5] [5] R. M. Wald, *Black hole in a uniform magnetic field*, Phys. Rev. D **10**, 1680 (1974). doi:10.1103/PhysRevD.10.1680.

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